

Matthew McCaughan

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Summary:

- Software Engineer with experience developing machine learning applications, AI-integrated systems, and database systems.
- Proficient in Python, Java, C++, and SQL.
- Projects include an interactive neural network-based digit recognition system, an AI chatbot leveraging RAG with live data injection, and a JDBC-based student database query platform.

Education:

Bachelor of Science, Computer Science, May 2025

Rutgers University, New Brunswick, NJ

Relevant coursework:

- Data Structures
- Algorithms
- Computer Architecture
- Databases
- Software Methodology
- Artificial Intelligence
- Machine Learning
- Computer Imaging & Multimedia

Skills:

- **Programming:** Python, Java, SQL, C++, Typescript
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, RAG
- **Data & Visualization:** Pandas, NumPy, Matplotlib, OpenCV, Pillow
- **GUI & Apps:** Android, JavaFX, Tkinter

Projects:

Project files available on [GitHub](#)

- **Interactive Handwritten Digit Recognition (LeNet-5 CNN)**
Python, PyTorch, OpenCV, Tkinter, NumPy, Matplotlib, Scikit-Learn

- Designed and implemented an **end-to-end machine learning application** that trains and deploys a **LeNet-5 convolutional neural network** for handwritten digit classification on the MNIST dataset (**~99% accuracy**).
 - Developed an **interactive GUI drawing interface with tkinter** allowing users to sketch digits and receive **real-time model predictions** through model inference.
 - Built an **image preprocessing pipeline** using OpenCV (thresholding, centering, normalization) to convert user-drawn input into model-compatible tensors.
 - **Structured the project as a modular ML system** including model training, evaluation, confusion matrix analysis, checkpointing, and an integrated user interface.
- **AI-Powered Interactive Portfolio & Resume ChatBot**
Gemini API, Github REST API, React, TypeScript, Vite, RAG Principles
 - Built and deployed a personal portfolio website to GitHub Pages using **React, TypeScript, and Vite**. Check it out at [my website!](#)
 - Integrated an LLM chatbot using the **Google Gemini API** allowing conversation history and responding with contextually accurate information about skills, projects, and experience.
 - Implemented a **RAG** pipeline that injects live **GitHub REST API** feeds and static resume information into the model's context when prompted.
 - Automated deployment via **GitHub Actions CI/CD**, implementing secure secret management for API credentials and enabling continuous delivery for every push on GitHub Pages.
 - **Student Database Query Tool:**
Python, Java, MySQL, JDBC API
 - **Designed and implemented a comprehensive Java-based student information query system** using JDBC, enabling interactive searches by GPA thresholds, major/minor, course enrollment, and department statistics.
 - **Built a SQL-integrated backend via JDBC integration** to dynamically query and aggregate student records (e.g., GPA calculations, major/minor listings, course grade breakdowns) from a relational database, supporting live user interaction in the command line.
 - **Developed reusable SQL logic and modular Java components** for handling student metadata, reducing redundant code across multiple query paths while improving maintainability and scalability for unseen data.
 - **Optimized database access patterns** through use of prepared statements and efficient result handling for querying majors, minors, GPA, and enrollment statistics, minimizing redundant SQL calls and enhancing performance.